TEXORA · LECTURE SHEET

Methods and Styles of Printing

Machines that transfer paste — block, roller, screen, rotary, transfer, digital — kept strictly separate from the chemical styles: direct, discharge, resist and burn-out.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 9 · LEVEL 3 TERM 1

1. Two independent classifications

Students confuse methods and styles because both lists describe printing. **Methods** answer *how is paste transferred?* — the machine. **Styles** answer *how is the design's colour pattern produced chemically?* — direct, discharge, resist. Any method can execute most styles; a flat-screen machine can print a discharge style, and a roller machine can print a resist style.

2. Methods of printing

METHOD	PRINCIPLE	STRENGTHS / LIMITS
Block	Hand-carved block dipped in paste, pressed on fabric	Craft/artisan; negligible capital; very slow
Roller (engraved copper)	Each colour = one engraved roller fed by a furnisher roller from a paste trough; fabric pressed against the set	Very high speed, fine lines; engraving costly — long runs only
Flat-bed screen	Flat screens (one per colour) lowered in sequence; squeegee strokes paste through mesh (Experiment 6 builds the screen)	Flexible design size, sampling-friendly; slower than rotary
Rotary screen	Seamless nickel cylinder screens rotate continuously over the moving fabric; paste fed inside, squeegee blade forces it out	The production standard: speed + screen flexibility
Transfer	Design printed on paper with disperse dyes, then heat-pressed (180–210 °C) so dye sublimes into polyester	Photographic quality on polyester; useless on cotton (no sublimation affinity)

Digital inkjet	Piezo heads jet dye inks (reactive, acid, disperse, pigment) drop-on-demand; fabric pre-treated, then steamed/washed as usual	No screens, unlimited colours, short runs; ink and machine cost, slower bulk speed
Flock / special	Adhesive printed, short fibres applied electrostatically (flock); also foil, puff, glitter pastes	Texture effects beyond colour

3. Styles of printing

STYLE	HOW THE PATTERN ARISES	NOTES
Direct	Colorant printed straight onto white or pale ground in the design areas	The default style; white ground stays unprinted
Discharge	Fabric first dyed with dischargeable dyes; print paste contains reducing agent (sodium formaldehyde sulphoxylate) that destroys ground dye where printed — white discharge, or coloured discharge if an illuminating dye stable to the reducer is included	Sharp designs on deep grounds impossible by direct printing; ground dye must be dischargeable
Resist	A resist (wax, resin, chemical agent such as alkali/acid or dye-blocking salts) is printed first and prevents dye uptake/fixation when the fabric is subsequently dyed or over-printed	Batik is the wax archetype; chemical resists are standard in reactive printing (e.g. printing acid-resist under reactive ground)
Burn-out (devoré)	Paste with acid (e.g. sodium hydrogen sulphate) carbonises and removes one fibre of a blend (cellulose from polyester/cotton) in the design areas	Transparent patterned effects; printed areas lose the cellulose entirely
Warp / ikat	The warp yarn is printed (or resist-dyed) before weaving	Soft-edged woven designs

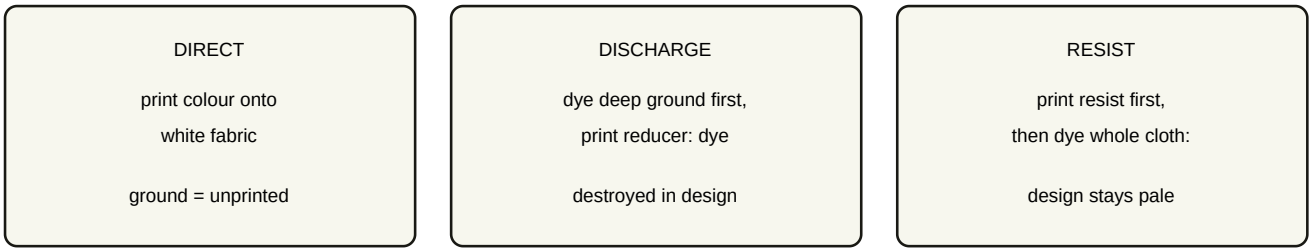


Fig. 1 — The three fundamental styles compared: where colour is added or prevented.

4. Matching method, style and fibre — worked logic

Deep navy ground with a fine white floral: **discharge style** (direct white pigment would sit dull and stiff; discharge gives a true white) on **rotary screen** for production. Polyester sportswear with a photographic gradient: **transfer method**, direct style. Cotton voile with sheer patterned panels: **burn-out style** on polyester/cotton. A 20-metre fashion sample in 12 colours: **digital inkjet** — twelve screens would never pay back. The reasoning always runs: effect wanted → style → fibre/dye system → method the run length justifies.

5. Glossary — key terms

- **Illuminating dye:** discharge-stable dye included in a discharge paste to colour the discharged area.
- **Furnisher roller:** roller feeding paste to an engraved printing roller.
- **Devoré:** burn-out printing on blends.
- **Repeat:** the unit length of design that tiles the fabric (roller circumference or screen length).

6. Key points

- Method = transfer machinery; style = chemical route to the pattern. Never use the words interchangeably.
- Discharge needs a dischargeable ground; resist needs dyeing after printing; transfer needs polyester and sublimable dyes.

7. Summary

Seven transfer methods and five pattern styles combine into the practical printing repertoire. Classification discipline — machine versus chemistry — lets a technologist specify any printed effect in one unambiguous sentence, which is the professional habit this topic exists to build.

8. Practice

1. Specify method + style + dye system for: (a) white polka dots on a deep red cotton ground, 5,000 m; (b) floral polyester scarf with photographic shading, 200 pieces; (c) sheer stripe effect on polyester/cotton shirting.
2. Why can transfer printing not be used on cotton, in terms of dye–fibre forces from Topic 2?
3. A discharge print shows a brownish halo around the design. What has happened chemically?

Answer key

1. (a) Rotary screen, discharge style (white discharge), ground dyed with dischargeable reactive/azo dyes. (b) Transfer printing (direct style) with disperse dyes. (c) Flat or rotary screen, burn-out style — acid paste removes cotton leaving polyester sheer stripes.
2. Transfer relies on the dye subliming and dissolving into the fibre as a solid solution — disperse dyes do this in hydrophobic polyester. Cotton offers no such phase: sublimed dye has no affinity for hydrophilic cellulose and simply re-deposits loosely, washing off.
3. Partial discharge at the design edge — reducer diffused beyond the print boundary and incompletely destroyed the ground dye, leaving its coloured reduction intermediates. Correct paste viscosity/reducer dose and steaming control.